

ERD Practice 1 (not graded)

Before working on the EER assignment, I strongly suggest you practice using the following example. The solution is appended on this page. We will also go over this practice question in our Zoom meeting. *(Please note that this is for your practice purpose, and will not be graded. You are not required to submit the answer to this practice.)*

Question:

- Wally would like to track the following information about its customers: Name, Address, City, State, Zip Code, Telephone, Date of Birth, and Primary Language.
- Customers place orders through a branch.
- The following information about each branch needs to be recorded: Branch Number, Location (Address, City, State, Zip Code), and Square Footage.
- While a customer may place many orders, he/she does not always have to order through the same branch all the time.
- Similarly, while a branch typically accepts orders from multiple customers, it may not receive any order from customers at a given time.
- The following information about each order needs to be recorded: Order Date and Credit Authorization Status.
- Customers may have multiple accounts, although they may also have no accounts.
- The following information needs to be recorded about accounts: Balance, Last payment date, Last payment amount, and Type.
- A branch may sell all items or may only sell certain items. Items may be sold by one or more branches.
- We would like to record the following about each item: Description, Color, Size, Pattern, and Type.
- Orders are composed of one or more items.
- An item can be composed of multiple items; for example, a dining room wallcovering set (item 20) may consist of wallpaper (item 22) and borders (item 23).
- In addition to fulfilling orders for customers, Wally's branches also perform other services to their customers.
- A branch may serve many customers but very occasionally does not serve any customers. A customer must be served by at least one branch but can choose different branches.
- Wally employs 56 employees and would like to track the following information about employees: Name, Address (Street, City, State, Zip Code), Telephone, date of Hire, Title, Salary, Skill, and Age.
- Each employee works in one and only one branch.
- Each employee may have one or more dependents. We wish to record the name of the dependent as well as the age and relationship.

- Employees can have one or more skills.

Follow the suggested steps below to develop an ERD.

1. Identify the entity types
2. Identify the relationship types and assert their degree
3. Assert the cardinality ratios and participation constraints (total versus partial participation)
4. Identify the attribute types and assert whether they are simple or composite, single- or multi-valued, derived or not
5. Link each attribute type to an entity type or a relationship type
6. Denote the key attribute type(s) of each entity type
7. Identify the weak entity types and their partial keys

Video Explanations:

Solution (*note: to achieve the best learning outcome, please look at the solution only after you have tried to solve it yourself*)

